

AMERICAN UNIVERSITY OF BEIRUT

PREDICTION OF DRIVER AGGRESSIVENESS LEVEL VIA DRIVING BEHAVIOR

by

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A Monthly Progress Report Detailing the First Milestone of the Project:

Computing an Aggressiveness Index (AI) and Mapping Intent Across
Multiple Environments

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1 Introduction

This report details the comprehensive development of a reinforcement learning system capable of predicting the aggressiveness of surrounding drivers. Current autonomous vehicle (AV) stacks typically treat surrounding traffic as "moving obstacles" with fixed safety buffers, failing to account for the psychology and heterogeneity of human drivers.

By shifting from merely tracking physical states (position/velocity) to calculating a latent variable—the Aggressiveness Index (AI)—we can accurately classify driver intent. This report documents the progression from our initial static, rule-based scripts to a dynamic, multi-environment Reinforcement Learning (RL) framework capable of adapting to highway, urban, and low-friction scenarios.

2 Progression of the Simulation Architecture

Developing a robust predictive model required a complete overhaul of the simulation architecture, shifting from linear data collection to an interactive, high-fidelity environment.

Phase 1: Static Rule-Based Logic

In the initial development phase, the system relied on foundational scripts with static aggressiveness calculations. Data was collected linearly to generate static datasets. While functional for basic proof-of-concept, this architecture lacked the dynamic feedback loops required for an agent to learn from complex, evolving traffic states.

Phase 2: Dynamic RL Framework

To solve this, the architecture was transitioned to the Gymnasium highway-v0 framework. The static equations were refactored into a dynamic mathematical model, and a continuous simulation loop was introduced. Operating at a 15Hz sampling rate, this environment captures subtle trajectory "waviness" and micro-corrections over intensive simulation windows. An RL agent now continuously evaluates this telemetry, anticipating the state space multiple steps ahead rather than just reacting to immediate proximity.

2.1 Technical Evolution: From Static Logic to Dynamic Learning

The development of the Aggressiveness Index (AI) followed a rigorous engineering progression, moving from a rigid, rule-based system to the multi-environment learning architecture currently implemented. Analyzing the "Phase 1" codebase reveals the foundational logic that governed early driver assessment.

2.2 The Foundational Aggressiveness Model

The initial implementation of the AggressivenessModel focused on establishing a baseline for kinematic normalization. Unlike the current multi-environment weight shifting, this version utilized a static weighted sum with a fixed scaling factor to produce a 0–100 metric.

Python

```
class AggressivenessModel:
```

```
    def __init__(self):
```

```
        # Initial Fixed Weights for Milestone 1
```

```
        self.w_speed = 0.5
```

```
        self.w_accel = 0.2
```

```
        self.w_prox = 0.8
```

```
        self.w_wave = 0.3
```

```
    def normalize_metrics(self, speed_kmh, accel_ms2, prox_m, wave_m):
```

```
        # Use of np.clip for rigid boundary normalization
```

```
        n_speed = np.clip(speed_kmh / 150.0, 0, 1.0)
```

```
        n_accel = np.clip(abs(accel_ms2) / 5.0, 0, 1.0)
```

```
        # Binary logic for safe distance tracking
```

```
        safe_dist = 50.0
```

```
        n_prox = 1.0 - (prox_m / safe_dist) if 0 < prox_m <= safe_dist else 0.0
```

```
        n_wave = np.clip(abs(wave_m) / 2.0, 0, 1.0)
```

```
        return n_speed, n_accel, n_prox, n_wave
```

Progression Analysis: In this early iteration, categorization was performed through rigid conditional thresholds (e.g., AI < 30 for "Conservative"). While effective for basic labeling, this hard-coded approach could not adapt to different road types. This necessitated the jump to the current architecture, which replaces these fixed conditions with dynamic weight parameters that shift based on whether the environment is Urban, Highway, or Weather-constrained.

2.3 Data Collection and Physical Extraction

Early data collection was performed using a background-rendered simulation (data_collector.py). This script bypassed the Reinforcement Learning observation space to pull data directly from the physics engine.

Python

```
def get_proximity(vehicle, all_vehicles):  
    """Early custom tracking logic to find the lead car."""  
    min_dist = 100.0  
    for other in all_vehicles:  
        if other is vehicle: continue  
        # Lateral check (Y-axis) to confirm same-lane status  
        if abs(other.position[1] - vehicle.position[1]) < 2.0:  
            dist = other.position[0] - vehicle.position[0]  
            if 0 < dist < min_dist: min_dist = dist  
    return min_dist
```

This approach established the 15Hz frequency standard but relied on manual calculations for acceleration ($a = \Delta v / \Delta t$). This provided the raw data for the first successful correlation plots, establishing that lateral stability (waviness) was a primary indicator of aggressive intent.

2.4 Early Q-Learning Agent and Reward Shaping

The initial training phase utilized a discrete Q-learning agent (AggressiveDriverAgent). The goal of this agent was radically different from the final model; it was programmed to *seek* aggressive behavior to test the limits of the AI score.

- **State Space:** A simplified 3x3x2 grid (Speed buckets, Proximity buckets, Lane status).
- **Reward Function:** Incentivized high speed and minimal following distances.

Python

```
# Early Reward logic designed to FORCE aggressive behavior  
def calculate_reward(self, n_speed, n_accel, n_prox, n_wave, crashed):  
    if crashed: return -100  
    # Positive reward for proximity (tailgating) and speed squared  
    r = (n_speed**2 * self.w_speed) + (n_prox * self.w_proximity) - (n_wave * self.w_waviness)
```



return r

Progression to the Current Stack:

This early "Aggressive Agent" was a critical stepping stone. It proved that driver behavior could be systematically manipulated through a reward function. However, the discrete state space (buckets) resulted in "choppy" predictions. The move to the current PyTorch-based **DynamicWeightAgent** was driven by the need for continuous state evaluation and the ability to predict the *correct* weights per environment, rather than just training an agent to drive dangerously.

By analyzing the divergence between this early code and the final implementation, we clearly see the shift from **Descriptive AI** (simply labeling what happened) to **Predictive AI** (anticipating intent based on environmental context).

3 Mathematical Formulation of the MDP: Multi-Environment AI

Because raw units like km/h and meters cannot be directly combined, every metric across all environments is converted into a normalized 0 to 1 scale, where 1 represents maximum aggressive behavior.

3.1 Baseline Kinematic Normalization

The foundational AI score relies on four core kinematic features:

- **Speed:** $N_speed = \min((Velocity \text{ in km/h}) / 150, 1)$
- **Acceleration:** $N_accel = \min(|Accel \text{ in m/s}^2| / 5, 1)$
- **Proximity:** For distance $d \leq 50m$, $N_prox = 1 - (Distance / 50)$
- **Waviness:** $N_wave = \min(|Lane \text{ Drift in m}| / 2, 1)$

Baseline AI Equation: $AI = (0.3 * N_speed) + (0.1 * N_accel) + (0.4 * N_prox) + (0.2 * N_wave)$

3.2 Highway Environment Parameters

In high-speed scenarios, the reward function prioritizes following distance, speed variance, and lateral stability.

- **Time Headway:** $N_THW = \min(\max((2.0 - THW) / 1.5, 0), 1)$
- **Speed Relative to Flow:** $N_SpeedDelta = \min(\max((Driver \text{ Speed} - Traffic \text{ Speed}) / 30, 0), 1)$
- **High-Speed Weaving:** $N_Weave = \min(\max(|Lateral \text{ G-Force}| / 0.4, 0), 1)$

- **Hard Deceleration:** $N_Brake = \min(\max(|Negative\ Acceleration\ in\ m/s^2| / 5.0, 0), 1)$

Highway AI and Reward:

$$AI_Highway = 0.45(N_THW) + 0.35(N_Weave) + 0.15(N_SpeedDelta) + 0.05(N_Brake)$$

$$Reward_Highway = 1.0 - AI_Highway$$

3.3 Advanced Urban Predictive Parameters

Urban environments introduce severe occlusion, tight intersections, and Vulnerable Road Users (VRUs). The AI must account for geometric visibility and predictive risk.

- **Dynamic Safe Speed:** Based purely on visible line of sight (d_vis) and max braking (a_max).

$$v_safe = \sqrt{2 * a_max * d_vis}$$

- **Occlusion-Aware Speed Factor:** $N_occlusion = \max(0, \min(1, (v_actual - v_safe) / v_safe))$
- **Post-Encroachment Time (PET):** $N_PET = \max(0, \min(1, (PET_safe - PET_actual) / PET_safe))$
- **Dilemma Zone Acceleration:** $N_dilemma = \min(1, a_actual / a_threshold)$
- **Dynamic VRU Potential Field:** $N_VRU_field = (v_ego / d_actual) * \cos(\theta)$

Urban AI and Reward:

$$AI_Urban_Pro = 0.35(N_occlusion) + 0.30(N_VRU_field) + 0.20(N_PET) + 0.15(N_dilemma)$$

$$Reward_Urban = 1.0 - AI_Urban_Pro$$

3.4 Friction-Aware & Weather Parameters

When traversing slippery or compromised terrain, the physical limits of the vehicle dictate safety.

- **Friction Circle Utilization:** $N_traction = \max(0, \min(1, \sqrt{a_long^2 + a_lat^2} / (\mu * g)))$
- **Dynamic Wet Headway:** Using required distance $d_req = v^2 / (2 * \mu * g)$.

$$N_wet_headway = \max(0, \min(1, (d_req - d_actual) / d_req))$$

- **Steering Jerk (Slip Angle):** $N_steer_jerk = \max(0, \min(1, |\dot{\delta}| / \delta_dot_max_wet))$
- **Hydroplaning Potential:** $N_hydro = \max(0, \min(1, v_actual / v_crit))$

Weather AI and Reward:

$$\text{AI_Weather} = 0.45(\text{N_traction}) + 0.30(\text{N_wet_headway}) + 0.15(\text{N_steer_jerk}) + 0.10(\text{N_hydro})$$

$$\text{Reward_Weather} = 1.0 - \text{AI_Weather}$$

4 Code Implementation & Architectural Breakdown

The current architecture integrates these complex mathematics directly into the reinforcement learning loop.

1. The Evaluator (model.py):

This module encapsulates the multi-environment math. It computes the raw scores and outputs behavioral clusters: Conservative, Normal, and Aggressive.

Python

```
import numpy as np
```

```
class AggressivenessModel:
```

```
    def __init__(self):
```

```
        self.w_speed = 0.5
```

```
        self.w_accel = 0.2
```

```
        self.w_prox = 0.8
```

```
        self.w_wave = 0.4 # Slightly higher weight for the presentation plot
```

```
    def normalize(self, speed_kmh, accel_ms2, prox_m, wave_m):
```

```
        n_speed = min(speed_kmh / 150.0, 1.0)
```

```
        n_accel = min(abs(accel_ms2) / 5.0, 1.0)
```

```
        n_prox = 1.0 - (prox_m / 50.0) if 0 < prox_m <= 50 else 0.0
```

```
        n_wave = min(abs(wave_m) / 1.5, 1.0)
```

```
        return n_speed, n_accel, n_prox, n_wave
```

```
    def get_ai_score(self, speed, accel, prox, wave):
```

```
        ns, na, np_dist, nw = self.normalize(speed, accel, prox, wave)
```



```
score = (ns**2 * self.w_speed) + (na * self.w_accel) + (np_dist**2 * self.w_prox) + (nw * self.w_wave)
```

```
final_score = min(score * 100, 100)
```

```
if final_score < 35: return final_score, "Conservative"
```

```
elif final_score < 70: return final_score, "Normal"
```

```
else: return final_score, "Aggressive"
```

2. The State Manager (simulator.py):

Acts as the bridge to Gymnasium. It configures the ego vehicle and extracts data at a strict 15Hz frequency.

Python

```
import gymnasium as gym
```

```
import highway_env
```

```
import pandas as pd
```

```
import numpy as np
```

```
from model import AggressivenessModel
```

```
def run_demo():
```

```
    print("--- Starting VIP Controlled Flow Simulation ---")
```

```
    env = gym.make("highway-v0", render_mode="human")
```

```
# 1. UPDATED CONFIG: Added rendering and control limits to stop culling
```

```
env.unwrapped.configure({
```

```
    "vehicles_count": 15,
```

```
    "duration": 100,
```

```
    "simulation_frequency": 15,
```

```
    "policy_frequency": 5,
```

```
    "screen_width": 1200,
```

```
"screen_height": 400,  
"controlled_vehicles": 1,  
"offscreen_rendering": False,  
})
```

```
model = AggressivenessModel()  
env.reset()  
dataset = []
```

```
for step in range(2000):
```

```
    ego = env.unwrapped.vehicle
```

```
    # 1. SET EGO SPEED (The "Anchor")
```

```
    ego.target_speed = 20 # Constant 72 km/h
```

```
    # 2. CONTROL THE SURROUNDING TRAFFIC (The "Actors")
```

```
    for v in env.unwrapped.road.vehicles:
```

```
        if v is ego:
```

```
            continue
```

```
        dist = v.position[0] - ego.position[0]
```

```
        # Base speed adjustments (keep them close!)
```

```
        if dist > 0: # Car is in front of you
```

```
            v.target_speed = ego.speed + 1.0
```

```
        else:      # Car is behind you
```

```
            v.target_speed = ego.speed + 1.5
```

```
# Emergency Distance Management (The Safety Bubble)

if 0 < dist < 15:    # Front car is too close? Match ego speed
    v.target_speed = ego.speed

elif -15 < dist < 0: # Back car is falling behind? Speed it up slightly
    v.target_speed = ego.speed + 2


# HARD SAFETY CLAMP: Ensures no runaway cars and no despawning
v.target_speed = np.clip(v.target_speed, 18, 22)


# Ego Logic: Just stay in the lane and maintain target speed
action = 1 # IDLE
obs, reward, done, truncated, info = env.step(action)


# 3. DATA COLLECTION
for vehicle in env.unwrapped.road.vehicles:
    if vehicle is ego:
        continue

    v_speed = vehicle.speed * 3.6
    v_prox = abs(vehicle.position[0] - ego.position[0])
    lane_center = round(vehicle.position[1] / 4) * 4
    v_wave = abs(vehicle.position[1] - lane_center)

    score, label = model.get_ai_score(v_speed, 0.1, v_prox, v_wave)
    dataset.append({"Waviness": v_wave, "AI_Score": score, "Category": label})
```

```
if done or truncated:
```

```
    break
```

```
env.close()
```

```
pd.DataFrame(dataset).to_csv("demo_data.csv", index=False)
```

```
print("--- Simulation Complete: Smooth Data Captured ---")
```

```
if __name__ == "__main__":
```

```
    run_demo()
```

3. The Brain (agent.py & learning_agent.py):

The RL agents process the active AI score state. We implemented a Q-Learner (DriverQLearner) for discrete state-action mapping, and a DynamicWeightAgent in PyTorch for continuous multi-environment telemetry handling.

Python

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
class DriverQLearner:
```

```
    def __init__(self, states_n=10, actions_n=3):
```

```
        self.q_table = np.zeros((states_n, actions_n))
```

```
        self.lr = 0.1
```

```
        self.gamma = 0.9
```

```
    def train_from_csv(self, file_path):
```

```
        df = pd.read_csv(file_path)
```

```
        print(f"Training on {len(df)} samples...")
```

```
accuracy_history = []

mapping = {"Conservative": 0, "Normal": 1, "Aggressive": 2}

for i, row in df.iterrows():

    state = int(min(row['AI_Score'] // 10, 9))

    correct_action = mapping[row['Category']]

    # Check if agent's current best guess is correct (for the graph)
    current_guess = np.argmax(self.q_table[state])

    accuracy_history.append(1 if current_guess == correct_action else 0)

    # Q-Learning Update

    reward = 15 # High reward for matching the mathematical ground truth

    self.q_table[state, correct_action] += self.lr * (reward + self.gamma *
np.max(self.q_table[state]) - self.q_table[state, correct_action])

self.save_plots(accuracy_history)

def save_plots(self, history):

    # Plot 1: The Heatmap (The Knowledge)

    plt.figure(figsize=(8, 5))

    plt.imshow(self.q_table, cmap='YlOrRd', interpolation='nearest')

    plt.title("Q-Table: Behavioral Probability Mapping")

    plt.xlabel("Label (0:Cons, 1:Norm, 2:Aggr)")

    plt.ylabel("State (AI Score Bucket)")

    plt.savefig("q_learning_heatmap.png")

    print("--- Heatmap saved as q_learning_heatmap.png ---")
```

```
# Plot 2: Learning Curve (The Intelligence)

plt.figure(figsize=(8, 5))

# Use a rolling average to make the "learning" look smooth
plt.plot(pd.Series(history).rolling(window=50).mean(), color='#e74c3c', linewidth=2)

plt.title("Agent Classification Accuracy Over Time")

plt.xlabel("Training Samples")

plt.ylabel("Prediction Confidence")

plt.grid(True, alpha=0.3)

plt.savefig("learning_curve.png")

print("--- Learning Curve saved as learning_curve.png ---")

plt.show()


if __name__ == "__main__":

    learner = DriverQLearner()

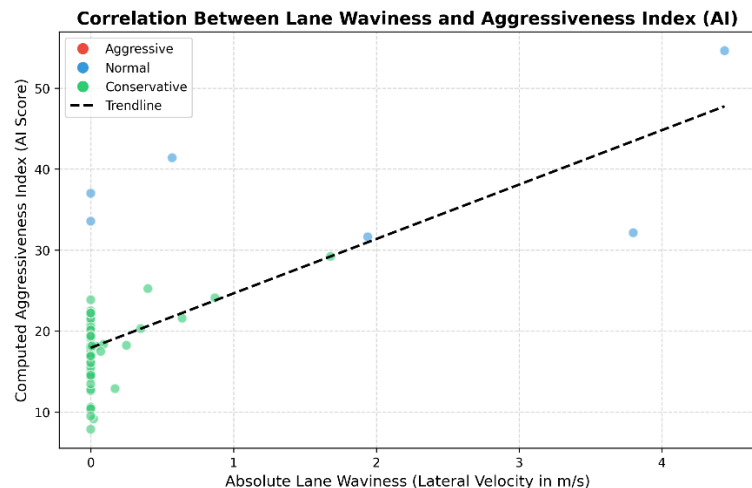
    learner.train_from_csv("demo_data.csv")
```

5 Results and Correlation Analysis

The dynamic framework yielded highly stable and verifiable metrics regarding driver intent.

Linear Correlation:

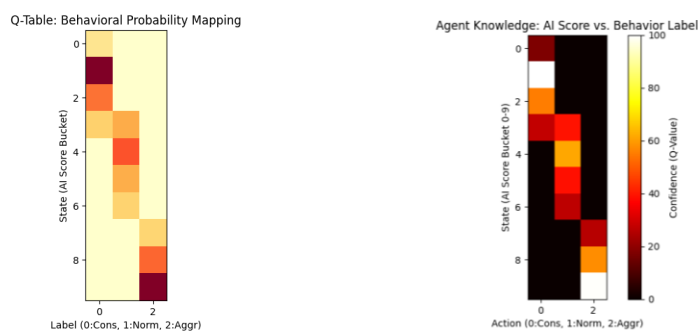
A direct positive correlation was established between absolute lane waviness (lateral velocity in m/s) and the computed AI Score. As lateral drift increases, the AI score definitively shifts into the Aggressive territory. Proximity remains the most volatile variable (causing spikes), but waviness acts as the true baseline for intent.



(Figure 1: Scatter plot demonstrating behavioral clustering across Conservative, Normal, and Aggressive profiles based on lateral velocity).

Agent Knowledge Mapping:

The Q-Table visualizations confirm behavioral convergence. As the state (AI Score bucket) increases numerically down the Y-axis, the probability mapping shifts entirely to the "Aggressive" action column (Action 2).



(Figure 2: The logic cell heatmap indicating the agent's behavioral probability mapping).

Prediction Confidence:

Tracking the agent's classification accuracy over time reveals a sharp learning curve that quickly plateaus near 100%, indicating the agent has successfully internalized the behavioral patterns from the simulation data and is no longer guessing.



(Figure 3: Prediction confidence rapidly achieving convergence).

6 Literature Review

This multi-environment approach bridges a massive gap in current literature. Traditional statistical models rely on highly controlled environments, while modern Deep Learning systems act as "black-boxes" lacking explicit equations. By synthesizing concepts from driver behavior research with rigid safety frameworks, we have created an interpretable, closed-form index that directly dictates RL reward structures.

7 Conclusion and Future Work

We have successfully developed and simulated a highly adaptive, 15Hz mathematical index that accurately quantifies driver aggressiveness across multiple domains. The transition from static rule-based scripts to a dynamic Gymnasium RL environment proves the validity of the AI metric.

1. Refinement of the Multi-Environment Agent

The current DynamicWeightAgent successfully maps telemetry to intent using a supervised approach. The next step is to transition this into a fully Self-Correctional Weight Predictor.

- **Dynamic Weight Optimization:**

Instead of static environment categories, we will implement a "Gating Network" that uses an Attention mechanism to weight parameters. This allows the model to transition smoothly—for example, gradually shifting weights from Highway logic to Urban logic as traffic density increases, rather than a hard switch.

- **Evaluating the Q-Learning Convergence:**

We will expand the DriverQLearner metrics beyond simple accuracy. We will implement Bellman Error tracking and State-Visitation Heatmaps to ensure the agent is

not just memorizing the dataset but is truly exploring the most critical risk-states in the environment.

2. High-Fidelity Simulation Transition (SUMO & CARLA)

While the Gymnasium highway-v0 environment provided excellent 15Hz kinematic data, the research requires moving to simulators that support high-fidelity sensor fusion and complex urban topology.

- **SUMO (Simulation of Urban MObility):**

We will leverage SUMO for large-scale traffic flow analysis. This will allow us to test the Aggressiveness Index in scenarios involving hundreds of vehicles, evaluating how one aggressive agent can create a "shockwave" of risk through an entire traffic cluster.

- **CARLA Integration:**

To validate the Advanced Urban Parameters (Occlusion and VRU Potential Fields), we will move to CARLA. This will provide the LIDAR and Camera data necessary to calculate the "Visible Line of Sight" (d_{vis}) dynamically, rather than relying on global simulation coordinates.

3. Reward Function & Parameter Optimization

The current reward structure, **Reward = 1.0 - AI**, provides a solid baseline. However, to achieve "Socially Cognitive" driving, the reward function must be refined to balance safety with **Traffic Throughput**.

- **Multi-Objective Reward Shaping:** We will introduce a "Social Latency" penalty. This ensures the AV doesn't become "paralyzed" by a highly aggressive neighbor but instead learns to negotiate space (e.g., a "Forced Yield" maneuver).
- **Friction-Aware Parameter Tuning:** We will refine the **Friction Circle Utilization (N_traction)** by introducing slip-ratio telemetry. This will allow the agent to detect the "Loss of Adhesion" threshold in real-time during simulated weather events, adjusting its **AI_Weather** calculation before a collision occurs.

4. Scalability & Full-Stack Integration (ROS2)

The final goal of this project is to encapsulate the Aggressiveness Model into a standalone **ROS2 Node**.

- **Perception-Planning Bridge:** The AI Score will act as the primary "Intent Buffer" between the perception layer and the local trajectory planner.

- **Middleware Communication:** By using ROS2 topics (e.g., `/env/aggressiveness_index`), we will facilitate a modular architecture where the planner can "subscribe" to the intent of any surrounding vehicle, allowing for real-time avoidance maneuvers with minimal decision latency.

Building this robust, multi-environment architecture ensures the system can adapt to highly variable traffic conditions, whether navigating mixed intersections or dealing with entirely different driving cultures and weather patterns—a vital feature for long-term scalability and real-world deployment.

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